

A Spherical N-Series Area Law for Regular Refinement Families

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Abstract

This paper presents a spherical extension of the N-Series geometric cancellation framework. The central claim is that regular, scale-consistent spherical area approximations can exhibit an even-power error expansion under refinement. When this structure is present, fixed N-Series scale operators cancel successive error terms and raise the observed convergence order. Four tests are reported. Spherical caps, chordal spherical triangle patches, and latitude–longitude spherical grid cells all exhibit the clean order ladder 2, 4, 6, 8 for the raw, two-scale, three-scale, and four-scale approximations respectively. A perturbation test then identifies the regularity boundary: the raw second-order law remains robust under smooth mesh perturbation, while higher-order cancellation becomes sensitive to the loss of scale regularity. The result is a constructive high-order area-refinement principle for regular spherical refinement families, not a black-box method for arbitrary irregular meshes.

1 Introduction

The N-Series framework begins from a simple observation: many geometric approximations carry structured asymptotic error expansions. When the leading errors occur in even powers of the scale parameter, approximations at several scales can be combined to cancel those leading terms. In planar polygonal and perimeter problems, this produces the order ladder

$$2, 4, 6, 8, \dots$$

The purpose of this paper is to show that the same structure appears in spherical area approximation. The tests considered here are deliberately elementary but geometrically distinct:

- a spherical cap approximated by a geodesic boundary polygon;
- a right spherical triangle approximated by a chordal triangulated patch;
- a latitude–longitude spherical cell approximated by a chordal grid;
- a perturbed latitude–longitude grid used to test robustness.

The first three tests confirm the clean N-Series order ladder for regular refinement families. The fourth test shows that high-order cancellation has a real regularity boundary. This is important: the method should be stated as a high-order principle for regular, scale-consistent refinement families, not as a universal correction for arbitrary meshes.

2 N-Series Scale Operators

Let D be a spherical domain on the unit sphere. Let $A(D)$ denote its exact spherical area, and let $A_N(D)$ denote a geometric area approximation at scale N . The spherical N-Series model is that,

for suitable regular refinement families,

$$A_N(D) = A(D) + \frac{C_2(D)}{N^2} + \frac{C_4(D)}{N^4} + \frac{C_6(D)}{N^6} + \cdots.$$

The coefficients depend on the domain and on the approximation family, but the powers are even.

The two-scale N-Series operator is

$$\hat{A}_N^{(2)}(D) = \frac{4 A_{2N}(D) - A_N(D)}{3}.$$

If the expansion above holds, then $\hat{A}_N^{(2)}(D) = A(D) + O(N^{-4})$.

The three-scale operator is

$$\hat{A}_N^{(3)}(D) = \frac{1}{45} A_N(D) - \frac{4}{9} A_{2N}(D) + \frac{64}{45} A_{4N}(D),$$

and gives $\hat{A}_N^{(3)}(D) = A(D) + O(N^{-6})$.

The four-scale operator is

$$\hat{A}_N^{(4)}(D) = -\frac{1}{2835} A_N(D) + \frac{4}{135} A_{2N}(D) - \frac{64}{135} A_{4N}(D) + \frac{4096}{2835} A_{8N}(D),$$

and gives $\hat{A}_N^{(4)}(D) = A(D) + O(N^{-8})$.

Thus the method is not a new local formula for spherical area. It is a scale-cancellation operator acting on a family of geometric approximations.

Relation to classical extrapolation. The operators above are not new as numerical devices. With scale doubling $N, 2N, 4N, 8N$ and a purely even-power error expansion, the weights $(4, -1)/3$; $(1, -20, 64)/45$; and $(-1, 84, -1344, 4096)/2835$ are exactly the Richardson–Romberg extrapolation weights that annihilate the leading $N^{-2}, N^{-4}, \dots$ error modes; each weight set sums to one, as it must for a consistent area estimate. The contribution of this paper is therefore *not* the extrapolation arithmetic, which is classical [3]. It is the prior fact that makes that arithmetic applicable here: that regular, scale-consistent *spherical geometric* area families possess an even-power error expansion at all, and that this structure — together with its breakdown under loss of regularity — can be exhibited across geometrically distinct spherical domains. The novelty is geometric, not algorithmic.

3 Conjectural Area Law

The numerical evidence motivates the following statement.

Spherical N-Series Area Law (conjectural). Let D be a spherical domain and let $A_N(D)$ be a regular, scale-consistent geometric area approximation at scale N . If the refinement family preserves the same geometric structure across the scales $N, 2N, 4N, 8N$, then the area error may admit an expansion of the form

$$A_N(D) - A(D) = \frac{C_2(D)}{N^2} + \frac{C_4(D)}{N^4} + \frac{C_6(D)}{N^6} + \cdots.$$

Under this even-power expansion, the N-Series operators produce the successive order improvements

$$O(N^{-2}) \longrightarrow O(N^{-4}) \longrightarrow O(N^{-6}) \longrightarrow O(N^{-8}).$$

The word *regular* is essential. The perturbation study below shows that raw second-order convergence can survive mesh perturbation, but higher-order cancellation is sensitive to the loss of scale regularity.

4 Evidence I: Spherical Cap Area

For a spherical cap on the unit sphere with polar angle α , the exact area is

$$A_{\text{cap}} = 2\pi(1 - \cos \alpha).$$

The cap boundary was approximated by a geodesic polygon with N equally spaced boundary vertices. The spherical polygon area was computed from spherical angle excess. The final observed orders at $N = 256$ are shown in Table 1.

α	Raw	Two-scale	Three-scale	Four-scale
15°	1.99989583	3.99999065	6.00093925	7.99993479
30°	1.99996740	4.00041698	6.00000053	7.95025527 [†]
60°	2.00016296	4.00014936	6.00003853	7.99880645
75°	2.00023455	4.00028432	6.00028420	8.00026989

Table 1: Observed orders for spherical cap area at $N = 256$. [†]The 30° four-scale value (7.95) sits below the clean 8.0 seen elsewhere. At $N = 256$ the four-scale error is already near the round-off floor, so the order estimate $p_N = \log(e_N/e_{2N})/\log 2$ destabilizes through near-cancellation of tiny errors; this is an artifact of the order estimator, not a failure of cancellation, and is confirmed by re-running at higher `mpmath` precision.

The cap test confirms the N-Series ladder in the simplest spherical setting: a rotationally symmetric domain with an exact closed-form area.

Why even powers (cap): a sketch. For the rotationally symmetric cap the even-power expansion can be argued, not merely observed. By rotational symmetry every one of the N geodesic boundary segments contributes identically, so the polygonal area is

$$A_N = N g(h), \quad h = \frac{2\pi}{N},$$

where $g(h)$ is the (single) area functional of one segment of angular width h subtended at the pole. Each geodesic segment is symmetric under reflection about its own midpoint relative to the latitude arc it replaces, so the per-segment area *deficit* $g(h) - A_{\text{cap}}/N$ is an even function of h to all orders accessible by the construction. Hence

$$A_N - A_{\text{cap}} = N[g(h) - A_{\text{cap}}/N] = \kappa_2 h^2 + \kappa_4 h^4 + \cdots = \frac{(2\pi)^2 \kappa_2}{N^2} + \frac{(2\pi)^4 \kappa_4}{N^4} + \cdots,$$

an expansion in even powers of $1/N$, which is exactly the structure the operators of Section 2 require. Equivalently, summing a smooth segment functional over a closed, uniformly sampled symmetric loop is an Euler–Maclaurin sum whose odd-order boundary terms cancel by periodicity and reflection symmetry. This is a sketch: full rigor requires bounding the remainder of g , which we do not carry out here. It does, however, place the strongest of the four tests on a constructive footing rather than a purely numerical one, and it makes precise why *regularity* (here, exact rotational symmetry) is what generates the even-power ladder.

5 Evidence II: Chordal Spherical Triangle Area

The second test uses a right spherical triangle on the unit sphere with vertices

$$A = (0, 0, 1), \quad B = (1, 0, 0), \quad C = (\cos \beta, \sin \beta, 0).$$

For this triangle, the exact spherical area is $A_{\text{exact}} = \beta$, where β is measured in radians. The approximation uses a normalized barycentric grid. Each small reference triangle is mapped to the sphere and treated as a Euclidean chordal triangle in $\mathbb{R}^3$. The raw area A_N is the sum of the chordal triangle areas. At $N = 64$, the observed orders are shown in Table 2.

β	Raw	Two-scale	Three-scale	Four-scale
15°	1.99942072	3.99915618	5.99886117	7.99716985
30°	1.99938698	3.99908575	5.99880136	7.99761444
60°	1.99923969	3.99881849	5.99858915	7.99835976
90°	1.99885313	3.99827889	5.99785193	7.99808456

Table 2: Observed orders for chordal spherical triangle area at $N = 64$.

This confirms that the order ladder is not restricted to boundary polygons. It also appears in a triangulated spherical patch.

6 Evidence III: Latitude–Longitude Cell Area

For a latitude–longitude cell on the unit sphere, $\lambda_0 \leq \lambda \leq \lambda_1$, $\phi_0 \leq \phi \leq \phi_1$, the exact area is

$$A_{\text{cell}} = (\lambda_1 - \lambda_0)(\sin \phi_1 - \sin \phi_0).$$

The approximation uses a uniform latitude–longitude grid. Each parameter cell is split into two chordal triangles in $\mathbb{R}^3$, and the raw area is the sum of the Euclidean triangle areas. At $N = 64$, the observed orders are shown in Table 3.

Cell	Raw	Two-scale	Three-scale	Four-scale
equatorial $30^\circ \times 30^\circ$	1.99996836	3.99999637	5.99994223	7.99911861
midlatitude $30^\circ \times 30^\circ$	1.99998486	3.99996146	5.99998582	8.00085300
wide equatorial $60^\circ \times 30^\circ$	1.99992674	3.99992210	6.00000864	8.00036842
northern $15^\circ \times 60^\circ$	1.99994200	3.99996772	5.99996443	8.00007636

Table 3: Observed orders for latitude–longitude spherical cell area at $N = 64$.

This test is the most directly applied regular-grid example. It connects the N-Series area law to grid-cell geometry used in spherical mesh and geodesy settings.

7 Perturbation Robustness

The fourth test perturbs the interior nodes of the latitude–longitude cell. The boundary nodes are fixed, while the interior grid points are displaced by a smooth deterministic field. The perturbation

Perturbation	Raw	Two-scale	Three-scale	Four-scale
0.00	1.99996836	3.99999637	5.99994223	7.99911861
0.02	1.99997529	4.02004815	4.53899093	2.61899119
0.05	2.00001207	3.44420303	5.68235637	2.77050496
0.10	2.00014360	3.89815337	8.94059560	3.59334301
0.20	2.00066838	3.90830154	6.15480240	4.36890149

Table 4: Mean final-level observed orders for perturbed latitude–longitude cell area.

amplitude is measured as a fraction of the local grid spacing. Three seeds were tested at each perturbation level. The aggregate final-level mean orders are shown in Table 4.

The perturbation result is not a failure of the framework. It is the regularity boundary of the framework, and it must be read carefully. Two facts are robust. First, the raw second-order law survives every tested perturbation: the raw column holds at 2.0 throughout. Second, the two-scale operator remains broadly effective, degrading only mildly (it stays near 4.0 except at the strongest single perturbation).

Beyond the two-scale operator, however, the reported orders no longer measure a convergence rate at all, and should not be interpreted as one. A theoretical three-scale operator cannot exceed order 6, yet the perturbed three-scale column reports 8.94 at perturbation 0.10 and swings non-monotonically ($4.54 \rightarrow 5.68 \rightarrow 8.94 \rightarrow 6.15$). An “observed order” above an operator’s own ceiling is the diagnostic signature that the even-power error structure has been destroyed: once it is gone, the operator no longer cancels a leading mode, the residual error is unstructured, and the estimator $p_N = \log(e_N/e_{2N})/\log 2$ returns a meaningless ratio of two unstructured small numbers. The four-scale column behaves the same way, collapsing to values between 2.6 and 4.4 that reflect noise, not eighth-order behavior.

The correct statement is therefore sharper than “the higher operators are useful but variable.” It is: smooth perturbation preserves the raw second-order law and largely preserves the two-scale operator, but breaks the even-power expansion that the three- and four-scale operators depend on; past that point their reported orders are artifacts of the estimator and carry no convergence meaning. The clean order ladder is a property of regular, scale-consistent refinement families, and only of those.

8 Interpretation

The evidence supports three conclusions. First, regular spherical refinement families can exhibit the same even-power area structure already seen in planar geometric approximations. Second, the N-Series operators act exactly as expected when the refinement family is regular and scale-consistent: raw second-order area convergence is raised to fourth, sixth, and eighth order. Third, high-order cancellation is not mesh-independent. Perturbing the mesh does not destroy raw second-order convergence, but it can break the higher N-Series operators. Thus the clean order ladder should be associated with regular refinement families, not arbitrary irregular meshes. This refined interpretation is

$$\text{regular spherical refinement} \implies \text{even-power area error} \implies \text{N-Series cancellation},$$

with the understanding that irregular perturbations may weaken the higher operators.

9 Relation to the General N-Series Framework

The spherical area law extends the N-Series framework from planar perimeter and chord approximation to curved-surface area. The operator mechanism is unchanged: one computes the same geometric quantity at several scales and combines those values with fixed weights so that the leading error modes vanish. The result is significant because it shows that N-Series cancellation is not limited to approximating lengths or constants such as π . It also applies to area approximation on curved surfaces when the refinement family preserves sufficient structure across scales.

10 Conclusion

The spherical cap, chordal spherical triangle, and latitude–longitude cell tests all confirm the same N-Series order ladder for regular refinement families: 2, 4, 6, 8. The perturbation test clarifies the scope of the result. Raw second-order area convergence is robust under smooth perturbation, but high-order cancellation depends on mesh regularity and scale consistency. The conservative conclusion is that spherical N-Series cancellation provides a constructive high-order area-refinement method for regular spherical refinement families. It should not be claimed as a black-box correction for arbitrary irregular meshes. Within its regularity class, however, the evidence strongly supports a spherical extension of the N-Series geometric cancellation framework.

References

- [1] H. S. M. Coxeter, *Introduction to Geometry*. Wiley, 1969.
- [2] I. Todhunter, *Spherical Trigonometry*. Macmillan, 1886.
- [3] L. F. Richardson, “The approximate arithmetical solution by finite differences of physical problems,” *Philosophical Transactions of the Royal Society A*, 210, 1911, pp. 307–357.
- [4] D. L. Williamson, “The Evolution of Dynamical Cores for Global Atmospheric Models,” *Journal of the Meteorological Society of Japan*, 85B, 2007, pp. 241–269.

A Reproducibility Notes

The numerical results in this paper were generated by four standalone Python scripts. Each script computes exact spherical reference areas, constructs a geometric approximation at several refinement scales, applies the N-Series operators, and reports observed convergence orders.

A.1 Spherical Cap Test

Generated by `spherical_area_nseries_cap_test.py`. For a cap of polar angle α , the exact area is $A_{\text{cap}} = 2\pi(1 - \cos \alpha)$. The script approximates the cap boundary by a geodesic polygon with N equally spaced boundary vertices, computes the polygonal area using spherical angle excess, applies the two-, three-, and four-scale N-Series operators, and reports observed orders.

A.2 Chordal Spherical Triangle Test

Generated by `spherical_triangle_nseries_area_test.py`. The target is a right spherical triangle with exact area $A_{\text{exact}} = \beta$, where β is the longitude separation of the equatorial vertices. The approximation uses a normalized barycentric grid; each mapped triangle is treated as a Euclidean chordal triangle in $\mathbb{R}^3$, and the raw area is the sum of those chordal areas.

A.3 Latitude–Longitude Cell Test

Generated by `spherical_latlon_cell_nseries_area_test.py`. For a unit-sphere cell the exact area is $A_{\text{cell}} = (\lambda_1 - \lambda_0)(\sin \phi_1 - \sin \phi_0)$. The script approximates the cell by a uniform latitude–longitude grid, maps it to the unit sphere, splits each parameter cell into two chordal triangles, and sums the Euclidean triangle areas.

A.4 Perturbed Latitude–Longitude Cell Test

Generated by `spherical_perturbed_latlon_nseries_test.py`. This uses the same exact cell area as the regular test, but perturbs the interior grid nodes by a smooth deterministic field while keeping boundary nodes fixed. The perturbation amplitude is a fraction of the local grid spacing. The perturbation field is scale-consistent, so the approximations at $N, 2N, 4N, 8N$ sample the same underlying perturbed geometry.

A.5 Reported Outputs

Each script writes a text report, a CSV summary, and a compact $\text{\LaTeX}$ table. The reported convergence orders are computed from successive errors by

$$p_N = \frac{\log(e_N/e_{2N})}{\log 2},$$

where e_N is the absolute error at scale N . This convention gives $p_N \approx 2$ for second-order convergence, $p_N \approx 4$ for fourth-order, and similarly for the higher N-Series operators.

A.6 Precision and Scope

All reported tests used high-precision arithmetic through `mpmath`. The results are numerical evidence for the stated spherical N-Series area law. They do not constitute a proof for arbitrary spherical domains or arbitrary irregular meshes. The perturbation test is included specifically to show that the clean high-order ladder depends on regularity and scale-consistency of the refinement family.